

Practical ON Image Processing

TITLE: To implement the Fourier Transform of an image using Python
and visualize its frequency components.

**BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)**

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Theory

The Fourier Transform (FT) converts an image from the spatial domain to the frequency domain. Low-frequency components represent smooth regions and overall brightness. High-frequency components represent edges, fine details, and noise. The Discrete Fourier Transform (DFT) is computed using the Fast Fourier Transform (FFT) algorithm.

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread('/content/image.webp', 0)

f = np.fft.fft2(img)

fshift = np.fft.fftshift(f)

magnitude_spectrum = 20 * np.log(np.abs(fshift) + 1)

f_ishift = np.fft.ifftshift(fshift)
img_back = np.fft.ifft2(f_ishift)
img_back = np.abs(img_back)

plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis('off')
(np.float64(-0.5), np.float64(499.5), np.float64(279.5), np.float64(-0.5))
```

Original Image



```
plt.subplot(1,3,2)
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Magnitude Spectrum")
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(499.5), np.float64(279.5), np.float64(-0.5))
```

Magnitude Spectrum



```
plt.subplot(1,3,3)
plt.imshow(img_back, cmap='gray')
plt.title("Reconstructed Image")
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(499.5), np.float64(279.5), np.float64(-0.5))
```

Reconstructed Image



```
plt.tight_layout()
plt.show()
```